

Camila Villalba Heuer

Backend Engineer - Distributed Systems / AWS

Java • Spring Boot • NestJS • TypeScript

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Professional Summary

Backend Engineer with software engineering experience across fintech and business applications, focused on building reliable backend systems, REST APIs, database-driven solutions and distributed workflows. Experienced with Java, Spring Boot, TypeScript, NestJS, PostgreSQL and AWS, with hands-on work involving financial transfers, CVU management, external payment integrations, authentication, asynchronous processing, retries, idempotency and transactional consistency.

Transitioned into software engineering in 2025 after more than three years in Chemical and Nuclear Engineering, where I worked with complex systems, technical troubleshooting, critical processes, multidisciplinary teams and highly regulated environments. This background strengthened my analytical, structured and problem-solving approach to software engineering.

Technical Skills

Backend Development: Java, Spring Boot, Node.js, NestJS, JPA/Hibernate, Spring Cloud, Spring AI

Architecture & Design: Domain-Driven Design (DDD), Hexagonal Architecture, Layered Architecture, Modular Monoliths, Microservices, Event-Driven Architecture, Use Cases, Repository Pattern, API Integration

Distributed Systems: Asynchronous Processing, SNS/SQS, Webhooks, Polling, Retries, Idempotency, Reconciliation, Error Handling, Transactional Integrity, Optimistic Locking

Security: Spring Security, JWT, API Key Authentication, OAuth2 concepts, Role-Based Authorization, AWS Cognito

Testing: Jest, Supertest, JUnit, Mockito, Unit Testing, Integration Testing

Databases: PostgreSQL, MySQL, DynamoDB, SQL, Prisma, Database Modeling, Transactions

Cloud & Infrastructure: AWS, Cognito, SNS, SQS, DynamoDB, Redis, S3, Secrets Manager, CloudFormation, Docker, CI/CD

Frontend & Others Technologies: React, Vite, HTML, CSS, Python, Git, GitHub, Swagger/OpenAPI

Data & Engineering Tools: Advanced Excel, Excel Dashboards, Power BI, Data Cleaning, SQL-based Data Analysis

Professional Experience

Fullstack Developer — Helipagos Fintech

Jun 2026 – Aug 2026

- Developed backend services and REST APIs for a fintech platform using NestJS, TypeScript, Prisma and PostgreSQL/Amazon RDS, working on core financial operations including transfers and CVU management.
- Implemented and evolved transfer management endpoints, including transfer listing with scope-based filtering and detail endpoints with role- and scope-based access control.
- Transfer status handling: bank responses could remain unresolved after the initial transfer request. Implemented a controlled polling workflow for UNKNOWN and IN_PROGRESS states, with scheduled processing, retries and a maximum of six attempts. Transfers that remained unresolved were persisted with a requires_reconciliation state, while terminal UNKNOWN_FOREVER responses were routed directly to reconciliation.
- Refactored transfer execution use cases to handle different bank response states and ensure the appropriate state transition, completion or failure flow depending on the external response.
- Implemented CVU registration and alias modification functionality, including DTO validation, business rules and persistence logic.
- Applied idempotency, transactional integrity, validation and error-handling mechanisms across financial operations, with structured logging and automated unit tests for controllers, use cases, services, repositories and adapters.
- Worked with Amazon Cognito for application authentication and authorization, including user groups and scope configuration for human/admin users.
- Integrated with existing AWS event-driven workflows, working with SNS/SQS-based processing and transfer-related events, and investigated existing DynamoDB and Redis usage for balance management and caching.

- Participated in the analysis and modernization of legacy functionality, code reviews and technical discussions, contributing to maintainability, reliability and consistency across the platform.

Software Engineer & Digital Solutions — Independent Projects

Aug 2025 – Present

Dental Management System

- Developed a fullstack clinical management system using React, Vite, NestJS, TypeScript, Prisma and PostgreSQL.
- Built patient management workflows covering medical history, treatments, visits and odontogram functionality.
- Designed relational data models and business rules for managing structured clinical information.
- Developed a modular application architecture focused on maintainability and future scalability.

Microservice Integrator – Payment Processing

- Designed and developed a Spring Boot microservice integrating with the Helipagos payment API to create, query and cancel payments through a secure REST API.
- Implemented payment idempotency using external references and database constraints, ensuring duplicate requests do not create duplicate payments under concurrent execution.
- Handled concurrency and consistency with transactional operations and optimistic locking when updating payment states.
- Implemented webhook-based state synchronization and reconciliation, including fallback matching by external reference when the remote payment identifier could not be persisted successfully.
- Designed failure-handling and retry strategies for communication failures with the external payment provider, preserving local payment records for subsequent reconciliation.
- Implemented API Key authentication, secure password and API Key storage, centralized exception handling, DTO validation, Flyway database migrations and automated tests with JUnit and Mockito.
- Containerized the application and database with Docker and configured CI/CD for automated build, testing and deployment.

Java / Spring Boot Projects

- Built multiple backend applications with Java and Spring Boot, applying object-oriented and functional programming principles, REST API design, JPA/Hibernate, validation and database persistence.
- Developed a distributed e-commerce application using Spring Cloud, Eureka, API Gateway and OpenFeign, implementing service discovery, inter-service communication and centralized routing.
- Applied Resilience4j to handle failures between services and improve the resilience of distributed application flows.
- Implemented authentication and authorization with Spring Security and JWT, including protected endpoints and role-based access control.
- Developed a clinical management application for patients and medical histories, applying layered architecture, Spring Data JPA and relational persistence.
- Built desktop applications with Java Swing as part of the exploration of different application architectures and Java technologies.

Head of Maintenance & Operations – RA-10 Nuclear Reactor (CNEA)

Mar 2023 – Oct 2025

- Led maintenance and operations activities for the RA-10 nuclear research reactor, coordinating multidisciplinary teams in a highly regulated and safety-critical environment.
- Coordinated engineers and technicians across maintenance, technical findings and operational activities, establishing work methods, priorities and technical criteria.
- Analyzed technical documentation, including design specifications, P&IDs, and operation and maintenance manuals, verifying consistency between design requirements, documentation and physical installation.
- Investigated and classified technical findings, evaluated their impact on system functionality and safety, and worked with engineering teams to define corrective actions and resolve discrepancies between design and implementation.
- Developed and standardized technical procedures and work methodologies, improving consistency, traceability and reliability across maintenance and operational activities.
- Supported the reactor licensing and commissioning process by reviewing technical information, validating compliance with requirements and maintaining traceability between technical decisions, documentation and implementation.
- Applied structured problem-solving and risk-based decision making to complex technical issues, while training technical personnel and using Excel, Power BI and MySQL to structure, analyze and report technical information.

Education

Specialization in Technological Applications of Nuclear Energy

Balseiro Institute / UBA (2022)

Chemical Engineering (Environmental Orientation)

National University of Misiones (2015 – 2021)

Additional Training

- Functional analysis - UTN.BA
- AWS Cloud Practitioner Essentials — Completed. Certificate of completion obtained; AWS Certified Cloud Practitioner exam pending.
- AWS AI Practitioner — Completed. AWS Certified AI Practitioner exam pending.
- Additional self-directed training in Java, Spring Boot, Spring Security, Spring Cloud, Spring AI, Node.js, NestJS, TypeScript, SQL, databases, AWS and software architecture.

Languages

Spanish: Native

English: B1 - Intermediate